

G1 (replicate 1)

$$J_old = Q/(P1+P2-Q)$$

A (added criterion)

G2 (replicate 2)

$$J_new = Qc/(P1a+P2b-Qc)$$

$$J_new \leq J_old \text{ iff } c(P1+P2) \leq P1a+P2b$$